

Full Stack Development with MERN

Project Documentation format

1. Introduction

- **Project Title:** Medical Diagnosis Center by Service Portal (ServiceNow)

2. Project Overview

Purpose:

The Medical Diagnosis Center is a healthcare management application developed on the ServiceNow platform. The purpose of this project is to digitize the complete diagnostic center workflow by providing online patient registration, diagnostic test booking, appointment management, laboratory report management, payment tracking, and automated notifications through the Service Portal.

Features:

- Patient Registration and Profile Management.
- Diagnostic Test Booking.
- Real-Time Appointment Slot Validation.
- Appointment Approval Workflow.
- Laboratory Report View and Download.
- Payment Status Tracking.
- Email Notifications.
- Admin Dashboard for monitoring appointments and reports.

3. Architecture

Frontend (Service Portal)

The frontend is developed using ServiceNow Service Portal with custom pages and widgets.

Pages included:

- Home Page.
- Book a Test Page.
- My Bookings Page.
- My Reports Page.
- Appointment Confirmation Page.
- Admin Dashboard Page.

Backend (ServiceNow Platform)

The backend is implemented using ServiceNow platform features.

Backend Components

- Custom Tables.
- Business Rules.
- Flow Designer.
- Client Scripts.
- UI Policies.
- Notifications.
- ACLs (Access Control Rules).

Database:

The application uses ServiceNow Custom Tables.

Table Name	Purpose
Patient Table	Stores patient information.
Diagnostic Test Table	Stores diagnostic test details.
Appointment Table	Stores appointment booking details.
Report Table	Stores laboratory reports.
Payment Table	Stores payment transaction details.

4. Setup Instructions

Prerequisites:

- ServiceNow Personal Developer Instance (PDI).
- ServiceNow Studio.
- Service Portal Plugin Enabled.
- Administrator Access.

Installation Steps:

- Login to ServiceNow Developer Instance.
- Open Studio.
- Create a Scoped Application named Medical Diagnosis Center.
- Create custom tables.
- Add required fields in each table.
- Configure Service Portal pages and widgets.
- Configure Business Rules and Flow Designer.
- Create notifications and ACLs.
- Test the application.

5. Folder Structure

ServiceNow Application Structure

Module	Purpose
Tables	Stores patient, appointment, report, payment, and test data.
Service Portal	Contains pages, widgets, menus, and themes.
Flow Designer	Automates appointment workflow and notifications.
Business Rules	Performs server-side validations.
Client Scripts	Handles client-side validations.
Notifications	Sends email notifications.
ACLs	Controls user permissions.

6. Running the Application

Steps to Run

1. Login to ServiceNow.
2. Open the Medical Diagnosis Center Service Portal.
3. Register as a patient.
4. Book a diagnostic test.
5. Complete payment.
6. View appointment confirmation.
7. Download laboratory reports.

7. API Documentation

This project uses ServiceNow Table APIs and platform services.

Module	Operation
Patient Table	Create, Read, Update patient records.
Appointment Table	Create appointment records and update workflow status.
Report Table	Create and retrieve laboratory reports.
Payment Table	Create payment records and update payment status.
Diagnostic Test Table	Retrieve available diagnostic tests.

8. Authentication

Authentication Method

ServiceNow User Authentication is used.

Authorization

Role-Based Access Control (RBAC) is implemented.

Role	Permissions
Patient	Book tests, view appointments, download reports.
Lab Technician	Update report status and upload reports.
Administrator	Manage tests, appointments, payments, reports, and users.

9. User Interface

The application provides a responsive Service Portal interface.

Main Pages

- Home Page
- Book a Test Page
- My Bookings Page
- My Reports Page
- Appointment Confirmation Page
- Admin Dashboard Page

UI Features

- Search bar.
- Test category filters.
- Appointment cards.
- Report summary cards.
- Payment details.
- Responsive layout for desktop and mobile.

9. Testing

Testing Strategy

The application was tested using User Acceptance Testing (UAT) and Functional Testing.

Testing Modules

Module	Result
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Patient Registration	Passed
Appointment Booking	Passed
Slot Validation	Passed
Workflow Automation	Passed
Report Download	Passed
Payment Status	Passed
Email Notifications	Passed
Role-Based Access	Passed

11. Screenshots or Demo

- Provide screenshots or a link to a demo to showcase the application.

12. Known Issues

During development, a few issues were identified and resolved.

Issue	Resolution
Duplicate appointment booking.	Business Rule added to validate available slots.
Payment status not updating automatically.	Flow Designer updated payment status.
Report download permission issue.	ACL permissions configured correctly.
Email notification delay.	Notification trigger updated.

13. Future Enhancements

Future improvements planned for the Medical Diagnosis Center application include:

- Online payment gateway integration (UPI, Cards, Net Banking).
- Home sample collection scheduling.
- Doctor consultation booking.

- SMS and WhatsApp appointment reminders.
- QR Code-based report download.
- Patient health history dashboard.
- Analytics dashboard for revenue and appointment trends.
- Mobile-friendly Service Portal enhancements.